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REVIEW

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DTaP-IPV-HB-Hib vaccine (Hexaxim): an update 10 years after first licensure

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ABSTRACT

Introduction: Hexaxim® is fully liquid, hexavalent, combination vaccine that provides immunization against diphtheria, tetanus, pertussis (whooping cough), polio, hepatitis B, and invasive diseases caused by *Haemophilus influenzae* type b. Combination vaccines such as Hexaxim reduce the number of injections needed, improving both vaccination compliance and operational efficiency.

Areas covered: Safety and immunogenicity data were reviewed from >25 clinical trials involving approximately 7200 infants/toddlers, identified using PubMed searches to April 2023. These trials have evaluated a diverse range of primary series and booster schedules, including antibody persistence, co-administration of Hexaxim with other routine pediatric vaccines, and specific populations (born to Tdap-vaccinated women, preterm, and immunocompromised infants). Lastly, post-marketing surveillance and real-world effectiveness data were assessed.

Expert opinion: An extensive program of clinical development prior to licensure demonstrated favorable vaccine safety and good immunogenicity of each antigen, and Hexaxim was first approved for use in 2012. In the 10 years since licensure, Hexaxim has been adopted widely, with more than 180 million doses distributed worldwide. The widespread use of this hexavalent vaccine is a crucial tool in the ongoing and future control of six pediatric infectious diseases globally.

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Clinical trial; effectiveness; fully liquid; hexavalent; immunogenicity; pertussis; safety; surveillance

1. Introduction

Hexaxim® (DTaP-IPV-HB-Hib) is a combination vaccine that provides immunization against six pediatric diseases: diphtheria (D), tetanus (T), pertussis (whooping cough), polio, hepatitis B (HB), and invasive diseases caused by *Haemophilus influenzae* type b (Hib). It is also referred to as Hexacima®, Hexyon®, and Hexarium® depending on the geographical area. Diphtheria, tetanus, pertussis, and polio are serious diseases that can lead to severe complications or even death in young children [1–4]. Invasive diseases caused by *Haemophilus influenzae* type b include meningitis and pneumonia [5]. Hepatitis B is a viral infection that can lead to chronic liver disease and liver cancer [6,7]. Hexaxim contains diphtheria, tetanus, acellular pertussis (aP), inactivated poliovirus (IPV), HB, and Hib (purified capsular polysaccharide conjugated to tetanus toxoid [PRP~T]) antigens. The vaccine is based on previous tetravalent DTaP-IPV (Tetraxim) [8] and pentavalent DTaP-IPV//Hib (Pentaxim) [9] vaccines, which have been extensively studied and are used worldwide. Hexaxim also includes a recombinant hepatitis B surface antigen (HBsAg) produced in yeast and specifically developed for inclusion in this vaccine [10]. The composition of Hexaxim is shown in Table 1. Its manufacture follows strict quality control measures and its reliability ensures lot-to-lot consistency in terms of vaccine immunogenicity and safety. To meet country-specific requirements and public health needs, both vial and pre-filled syringe presentations are available.

The first marketing authorization was obtained in Peru in December 2012 (based on the European Medicines Agency [EMA] Article 58 procedure) [11]. Hexaxim was approved for use in the European Union (EU) in April 2013 [12] and is now approved in 120 countries worldwide (Sanofi, data on file). The vial presentation is pre-qualified by the World Health Organisation (WHO) [13]. Hexaxim was developed for use worldwide outside the U.S.A. The extensive clinical evaluation showed a consistent and good safety profile and demonstrated Hexaxim's ability to induce a robust immune response against each targeted pathogen, with high levels of protection observed in a wide range of clinical settings [14–16]. Since licensure, clinical evaluation has continued to support the safety profile and immunogenicity in different settings including in specific populations. From its first use in June 2013 and up to the cutoff date of April 2023, more than 180 million doses of Hexaxim have been distributed worldwide. Figure 1 shows the use of Hexaxim in clinical practices around the world between June 2013 and April 2023.

With 10 years of post-licensure experience with Hexaxim, this article reviews the breadth of evidence accumulated in clinical studies as well as in real-world settings, reflecting on the role of Hexaxim in ensuring equitable access to reliably safe and effective vaccines to contribute to disease control and polio eradication. Relevant data were identified using PubMed searches to April 2023.

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Article highlights

- Hexaxim is a fully liquid, hexavalent vaccine for primary and booster vaccination of infants and toddlers from 6 weeks of age against diphtheria, tetanus, pertussis, polio, hepatitis B, and invasive diseases caused by *Haemophilus influenzae* type b.
- Hexaxim was first approved in 2012, and an extensive program of clinical trials supports its use in a diverse range of primary series and booster schedules, including co-administration of with other routine pediatric vaccines and describes its use in specific populations (pre-term and immunocompromised infants).
- Antibody persistence has been demonstrated up to school age for all Hexaxim antigens, and persisting immune memory against hepatitis B has been shown to last at least 9-10 years.
- In a real-world study, vaccine effectiveness of the acellular pertussis antigens included in Hexaxim was demonstrated, with sustained protection until children were scheduled to receive a school-age booster.
- In the 10 years since licensure more than 180 million doses of Hexaxim have been distributed worldwide, and post-marketing surveillance has confirmed the favorable safety profile of Hexaxim following large-scale vaccination programs worldwide, which is in line with other similar multivalent pediatric vaccines.
- The reliability of Hexaxim manufacturing ensures lot-to-lot consistency in terms of vaccine immunogenicity and safety.
- The inclusion in Hexaxim of acellular pertussis and inactivated poliovirus antigens is aligned with modern pediatric vaccination strategies and helps to provide equitable access to the control of childhood diseases and guard against outbreaks.

2. Body

2.1. Clinical development of Hexaxim

Hexaxim benefits from Sanofi's strong legacy in the development of DTaP-IPV – containing vaccines and decades of experience with a tetravalent vaccine (Tetraxim [DTaP-IPV]) and pentavalent vaccine (Pentaxim [DTaP-IPV//Hib]) [8,9]. Hexaxim contains the well-established antigens used in Pentaxim (DTaP-IPV//Hib) plus an HBsAg, and is presented as a fully liquid, ready-to-use suspension for injection without preservative and adjuvanted with aluminum hydroxide [12].

Prior to its inclusion in Hexaxim, the HB antigen was evaluated in two randomized, controlled trials (RCTs) in adolescents and adults, which demonstrated its non-inferior immunogenicity and similar safety profile compared to a licensed standalone HB vaccine. The hexavalent combination vaccine, Hexaxim, including 10 µg of this HB antigen was then evaluated in a thorough international clinical development program, assessing its use in infants and toddlers in a diverse range of settings. More than 25 clinical trials have been completed, involving approximately 7200 infants/toddlers.

The following sections provide an overview of the immunogenicity and safety evaluations. All trials in healthy infants/toddlers are summarized in Table 2, and seroconversion rate (pertussis antigens: pertussis toxin [PT] and filamentous hemagglutinin [FHA])/seroprotection rate (other antigens) data are presented in Figure 2 (primary series) and Figure 3 (booster). Primary schedules evaluated include 6, 10, 14 weeks of age, 2, 3, 4 months of age, 2, 4, 6 months of age, and 3, 5 months of age, and a booster has been evaluated from 11 months of age through the second year of life, and seroconversion rate/seroprotection rate data are shown for each schedule in Supplementary Figures S1–4.

2.1.1. Immunogenicity

2.1.1.1. 6, 10, 14 weeks of age primary series and booster at 15–18 months of age.

The 6, 10, 14 week primary series and 15–18 months of age booster schedule was evaluated in one trial in South Africa [17,18] and a post-licensure trial in India [19] (Table 2).

In the trial in South Africa (Supplementary Figure S1), post-primary series seroprotection rates (anti-D, anti-T, anti-Polio1,2,3, anti-PRP, anti-HBs) were ≥95.4% in the group receiving Hexaxim (with no HB vaccine at birth) as well as in the control group (DTwP/Hib, HB, and OPV) and non-inferiority of immunogenicity was shown post-primary series for these Hexaxim antigens (in the cohort with no HB vaccine at birth) versus comparator vaccines (DTwP/Hib, HB, and OPV). Post-primary series, anti-HBs seroprotection rate (≥10 mIU/mL) was high whether a birth dose of HB vaccine was given (99.0%) or not (95.7%), however, anti-HBs geometric mean concentration was higher following a birth dose of HB vaccine (1913 mIU/mL versus 330 mIU/mL). The booster vaccination elicited a strong response in each group for all antigens (seroconversion rates [PT, FHA] >83% and seroprotection rates [other antigens] >90%) with no notable differences between groups.

The trial in India was non-comparative and showed high immunogenicity for each Hexaxim antigen following a 6, 10, 14 week primary vaccination series with a standalone HB vaccination at birth with >93% vaccine response rates (PT, FHA)/seroprotection rates (other antigens).

In conclusion, Hexaxim in the Expanded Program on Immunization (EPI) schedule, with or without hepatitis B vaccine at birth, is highly immunogenic compared to control vaccines.

2.1.1.2. 2, 3, 4 months of age primary series and booster at 11–17 months of age.

The 2, 3, 4 month primary series and booster schedule was evaluated in one trial in Turkey [20], and

Table 1. Hexaxim composition (per 0.5 mL dose).

Active ingredients	
Diphtheria toxoid	≥20 IU (30 Lf)†
Tetanus toxoid	≥40 IU (10 Lf)‡
Bordetella pertussis antigens	
Pertussis toxoid	25 µg
Filamentous hemagglutinin	25 µg
Inactivated poliovirus¶:	
Type 1 (Mahoney)†	29 D antigen units#
Type 2 (MEF-1)†	7 D antigen units#
Type 3 (Saukett)†	26 D antigen units#
Hepatitis B surface antigen*	10 µg
Haemophilus influenzae type b polysaccharide:	
(Polyribosylribitol phosphate)	12 µg
Conjugated to Tetanus protein	22–36 µg
Adjuvant	
Aluminum hydroxide, hydrated	0.6 mg (expressed as Al^{3+})

†As lower confidence limit (p = 0.95) and not less than 30 IU as mean value.
‡As lower confidence limit (p = 0.95).
¶Poliovirus antigens cultivated on Vero cells.
#These antigen quantities are strictly the same as those reported elsewhere as 40-8-32 D-antigen units for Type 1, Type 2, and Type 3, respectively, when measured by another suitable immunochemical method.
*Hepatitis B surface antigen produced from recombinant strain of the yeast *Hansenula polymorpha*.
Note: the vaccine may contain traces of glutaraldehyde, formaldehyde, neomycin, streptomycin, and polymyxin B.
IU, international units; Lf, limit of flocculation.

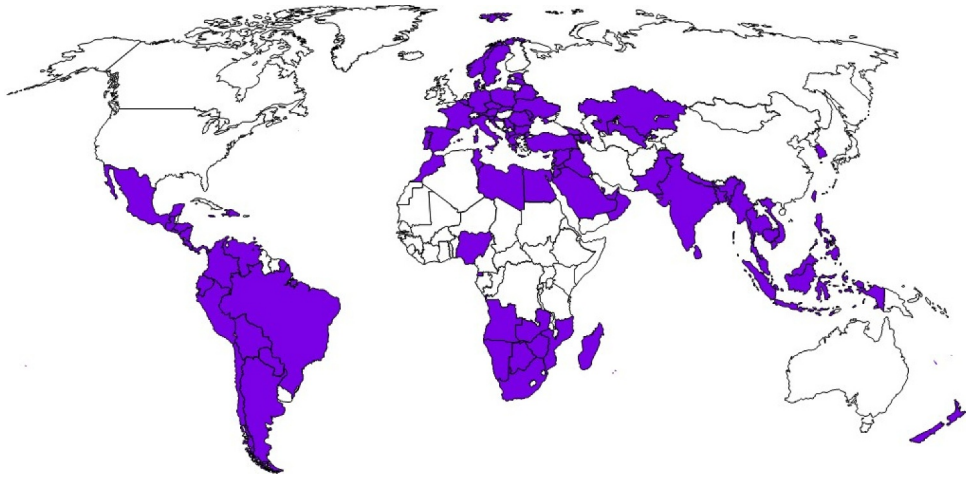


Figure 1. Hexaxim use in clinical practices around the world in the past decade (2013 – 2023).
Countries where Hexaxim has been used in clinical practices between June 2013 and April 2023 are shown in purple.
Depending on the country, Hexaxim may be funded/reimbursed as part of the National Immunization Program or available only as an out-of-pocket vaccine.

in three post-licensure trials in Germany and the Czech Republic [21], Vietnam (following a birth dose of HB vaccine) [22], and Finland [23,24] (Table 2 and Supplementary Figure S2). The EPI schedule starting at 6 weeks of age and the 2, 3, 4 month schedule are both considered to be the most challenging primary series schedules to achieve protective antibody titers as they start early in life (at a time when the immune system is not fully mature) and have only a 1-month interval between doses.

For the trial in Turkey, which compared Hexaxim to Pentaxim given concomitantly with standalone HB vaccine, non-inferiority was demonstrated post-primary series for the HB antigen of Hexaxim versus control HB vaccine in terms of seroprotection rate (≥10 mIU/mL) (94.0% for Hexaxim versus 96.1% for the HB vaccine). The immunogenicity of the other antigens was similar post-primary series for both Hexaxim and the comparator vaccine (Pentaxim) with seroconversion rates (PT, FHA) >81% and seroprotection rates (other antigens) >90%. Booster responses, with seroconversion rates (PT, FHA) >91% and seroprotection rates (other antigens) >97%, were high and similar for each antigen whether given after Hexaxim or Pentaxim and HB vaccine.

Of the three post-licensure trials, one was an RCT versus another hexavalent vaccine (Infanrix hexa [DTPa-HBV-IPV/Hib; GSK]) [21]. Primary series immune responses were high for all antigens and similar for both groups with vaccine response rates (PT, FHA) >94% and seroprotection rates (other antigens) >90%. Non-inferiority of the post-primary series immune response was demonstrated for all tested antigens (HB, PRP, PT, FHA). Descriptively, booster immunogenicity of Hexaxim was similar to Infanrix hexa for all antigens. The remaining two trials confirmed the good immunogenicity of Hexaxim with or without a dose of HB vaccine at birth.

2.1.1.3. 2, 4, 6 months of age primary series and booster at 12–24 months of age.

The 2, 4, 6 month primary series and 12–24 months of age booster schedule was evaluated in six trials in Mexico and Peru [25], Argentina [26,27], Mexico [28], Thailand [29], Peru [30], Colombia/Costa Rica [31] and in one post-licensure trial in South Korea [32] (Table 2 and Supplementary Figure S3). For the trial in Mexico and Peru, the primary objective was safety.

In the Argentina trial, non-inferior immunogenicity was shown post-primary series for each antigen versus Pentaxim and standalone HB vaccine (seroconversion rates [PT, FHA] ≥90% and seroprotection rates [other antigens] >94%). Similarly, non-inferiority was demonstrated post-primary series for all antigens versus Pentaxim and standalone HB vaccine (South Korea, seroconversion rates [PT, FHA] >89% and seroprotection rates [other antigens] >96%).

Non-inferiority of the post-primary immune response was demonstrated for all antigens versus Infanrix hexa (Colombia and Costa Rica: vaccine response rates [PT, FHA] >97% and seroprotection rates [other antigens] >94%). Non-inferiority to Infanrix hexa was demonstrated whenever tested in the trials in Mexico (anti-D), Thailand (anti-HBs, anti-PRP), and Peru (anti-HBs). Immunogenicity was descriptively similar for Hexaxim and the control vaccine for each antigen in all trials.

Hexaxim was highly immunogenic as a booster dose following a primary series of Hexaxim or Infanrix hexa (Mexico, Colombia/Costa Rica), and showed immunogenicity that was descriptively similar to Infanrix hexa when used as booster (Colombia/Costa Rica).

2.1.1.4. 3, 5 months of age primary series and booster at 11 months of age (2 + 1 schedule).

The 2-dose primary series schedule at 3 and 5 months of age with a booster at 11 months of age was evaluated in one post-licensure trial in Finland and Sweden [33] (Table 2 and Supplementary Figure S4), which showed non-inferiority versus Infanrix hexa for all antigens after the third dose in terms of seroprotection rates (≥85%) and vaccine response rates (≥98%).

2.1.1.5. Co-administration with other pediatric vaccines.

Co-administration with other pediatric vaccines was evaluated in several clinical trials (Table 2) showing similar immunogenicity whether the co-administered vaccine was given with Hexaxim or control vaccine(s), leading to approval for co-administration of Hexaxim with pneumococcal conjugate vaccines [21,23,29,31,33], measles-mumps-rubella (MMR) [18], and varicella containing vaccine [18] (and Sanofi, data on file),

Table 2. Summary of Hexaxim clinical trials (primary series, booster, and antibody persistence).

Trial	Country	Year	HB vaccine at birth	PS	B	Schedule	Control vaccine(s)	Co-administration with other vaccines	Total number of subjects (Subjects receiving Hexaxim)	Key findings	Reference
6-10-14 weeks primary series & booster											
A3L15	South Africa	2006-2007	No / Yes	✓		6-10-14 weeks	CombAct-Hib® (DTwP/Hib) + Engerix B (Hep B) + OPV	*None*	715 (243 without HB at birth, 137 with HB at birth)	Hexaxim immunogenicity was non-inferior to control group (without HB vaccine at birth) Hexaxim was well tolerated Hexaxim in the EPI schedule, with/without HB vaccine at birth, is highly immunogenic and safe	[17]
		2008-2009			✓	15-18 months	CombAct-Hib® (DTwP/Hib) + OPV	Trimovax (MMR) & Varilrix (Varicella) vaccines	567 (218 without HB at birth, 130 with HB at birth)	Hexaxim was highly immunogenic and well tolerated as booster Hexaxim can be given with MMR or varicella vaccines	[18]
A3L33	India	2014	Yes	✓		6-10-14 weeks	*None*	*None*	177 (177)	Hexaxim was highly immunogenic and well tolerated in Indian infants	[19]
2-3-4 months primary series and booster											
A3L10	Turkey	2006-2007	No	✓		2-3-4 months	Pentaxim + Engerix B (Hep B)	*None*	310 (155)	Hexaxim immunogenicity was non-inferior (tested only for Hep B) or similar to Pentaxim + HB vaccine Hexaxim was well tolerated, like Pentaxim + HB vaccine	[20]
A3L22		2007-2008			✓	15-18 months	*None*	*None*	254 (254)	Hexaxim was highly immunogenic and well tolerated as booster Hexaxim can be given as booster after primary series with Pentaxim + HB vaccine	[20]
A3L39	Germany Czech Rep	2014	No	✓		2-3-4 months	Infanrix hexa	Prevenar 13 (PCV13), RotaTeq (RV5)	529 (266)	Hexaxim immunogenicity was non-inferior (tested only for HB, PRP, PT, FHA) or similar to Infanrix hexa Hexaxim was well tolerated, like Infanrix hexa Hexaxim can be co-administered with PCV13 and rotavirus vaccine	[21]
A3L40		2014-2015			✓	11-15 months	Infanrix hexa	Prevenar 13 (PCV13)	464 (234)	Hexaxim was highly immunogenic and well tolerated as booster, like Infanrix hexa	[21]
A3L35	Vietnam	2015-2016	Yes	✓		2-3-4 months	*None*	*None*	354 (354)	Hexaxim was highly immunogenic (non-inferior to historical control for all antigens) and well tolerated in Vietnamese infants, when given as primary series or booster	[22]
		2016-2017			✓	16-17 months	*None*	*None*	349 (349)		
A3L43	Finland	2013	No	✓		2-3-4 months	*None*	Prevenar 13 (PCV13), RotaTeq (RV5) ± NeisVac-C (MenC)	350 (350)	Hexaxim immunogenicity was non-inferior (tested only for HB) or similar with or without concomitant MenC vaccine Hexaxim was well tolerated with or without concomitant MenC vaccine Hexaxim can be co-administered with MenC vaccines	[23]
		2014			✓	12 months	*None*	± Nimenrix (MenACWY)	312 (312)	Hexaxim immunogenicity was similar with or without concomitant MenACWY vaccine Hexaxim was well tolerated with or without concomitant MenACWY vaccine Hexaxim can be co-administered with MenACWY vaccines	[24]
2-4-6 months primary series & booster											
A3L04	Mexico Peru	2006-2008	No	✓		2-4-6 months	Tritanrix-HB/Hib (DTwP-HB/Hib) + OPV	*None*	2133 (1422)	In this large-scale safety study, Hexaxim was less reactogenic than the DTwP-combined vaccine Hexaxim was immunogenic for the HB component (subset)	[25]
A3L02	Argentina	2004-2005	No	✓		2-4-6 months	Pentaxim + Engerix B (HB)	*None*	624 (312)	Hexaxim immunogenicity was non-inferior (all antigens) to Pentaxim + HB vaccine Hexaxim was well tolerated, like Pentaxim + HB vaccine	[26]
A3L16		2006			✓	18 months	*None*	*None*	458 (None -Pentaxim booster)	Pentaxim was highly immunogenic and well tolerated as booster Pentaxim can be given after Primary Series with Hexaxim	[27]

(*Continued*)

Table 2. (Continued).

Trial	Country	Year	HB vaccine at birth	PS	B	Schedule	Control vaccine(s)	Co-administration with other vaccines	Total number of subjects (Subjects receiving Hexaxim)	Key findings	Reference
A3L31	South Korea	2014-2016	Yes	✓		2-4-6 months	Pentaxim + Euvax B (HB) (1, 6 months)	*None*	310 (153)	Hexaxim immunogenicity was non-inferior (all antigens) to Pentaxim + HB vaccine Hexaxim was well tolerated, like Pentaxim + HB vaccine Hexaxim was highly immunogenic and well tolerated in Korean infants	[32]
A3L11	Mexico	2006-2008	No	✓		2-4-6 months	Infanrix hexa	*None*	1189 (1022)	Lot-to-lot consistency of Hexaxim was demonstrated (all antigens) Hexaxim immunogenicity was non-inferior (tested only for diphtheria) or similar to Infanrix hexa Hexaxim was well tolerated, like Infanrix hexa	[28]
A3L21		2008-2009			✓	15-18 months		*None*	881 (768)	Hexaxim was highly immunogenic and well tolerated as booster. Hexaxim can be given after Primary Series with Infanrix hexa	[28]
A3L12	Thailand	2006-2007	Yes	✓		2-4-6 months	Infanrix hexa	Prevenar (PCV7)	412 (206)	Hexaxim immunogenicity was non-inferior (tested only for Hep B, PRP) or similar to Infanrix hexa Hexaxim was well tolerated, like Infanrix hexa	[29]
A3L17	Peru	2008-2009	No	✓		2-4-6 months	Infanrix hexa	*None*	266 (132)	Hexaxim immunogenicity was non-inferior (tested only for Hep B) or similar to Infanrix hexa Hexaxim was well tolerated, like Infanrix hexa	[30]
A3L24	Colombia Costa Rica	2010-2011	Yes	✓		2-4-6 months	Infanrix hexa	Prevenar (PCV7), Rotarix (RV1)	1376 (1031)	Lot-to-lot consistency of Hexaxim was demonstrated (all antigens) Hexaxim immunogenicity was non-inferior (all antigens) to Infanrix hexa Hexaxim was well tolerated, like Infanrix hexa Hexaxim can be co-administered with PCV7 or RV1	[31]
A3L27		2011-2012			✓	12-24 months	Infanrix hexa	Prevenar (PCV7)	1106 (691)	Hexaxim was highly immunogenic and well tolerated as booster, like Infanrix hexa Hexaxim can be given after Primary Series with Infanrix hexa	[31]
2-dose primary series & booster											
A3L38	Finland Sweden	2012-2013	No	✓		3-5 months	Infanrix hexa	Prevenar13 (PCV13) Rotarix (RV1) / RotaTeq (RV5)	550 (275)	Hexaxim immunogenicity post-primary series was high and non-inferior (all antigens except Polio 1 & 2) to Infanrix hexa Hexaxim was well tolerated, like Infanrix hexa Hexaxim can be co-administered with PCV13	[33]
		2013-2014			✓	11-12 months	Infanrix hexa	Prevenar13 (PCV13)	624 (312)	Hexaxim immunogenicity post-booster was high and non-inferior (all antigens) to Infanrix hexa. Hexaxim was well tolerated, like Infanrix hexa Hexaxim can be co-administered with PCV13	[33]
Mixed schedule											
A3L39	Spain	2014	Yes	✓		2-4-6 (Hexa-Penta-Hexa)	*None*	Prevenar13 (PCV13) RotaTeq (RV5) NeisVac-C (MenC)	265 (265)	Hexaxim was highly immunogenic and well-tolerated when used in a mixed hexavalent-pentavalent-hexavalent primary series Hexaxim can be co-administered with rotavirus vaccine (RV5) Hexaxim can be used in a sequential hexavalent-pentavalent hexavalent primary vaccination series followed by a pentavalent booster in coadministration with other common childhood vaccines	[34]
A3L40		2014-2015			✓	18 months	*None*	Prevenar13 (PCV13)	198 (None - Pentaxim booster)		[34]
A3L45	Russian Federation	2015-2016	Yes	✓		3-4.5-6 (Penta-Penta-Hexa)	*None*	*None*	100 (100)	Hexaxim was immunogenic and well tolerated when used at 6 months of age in infants previoulsy vaccinated with 2 doses of standalone HB vaccine (birth and 1 month of age) and 2 doses of pentavalent vaccine (3 and 4.5 months of age)	[80]
Antibody persistence											
A3L26 (Follow-up of A3L15)	South Africa	2010-2011	No / Yes	6-10-14 weeks & booster; persistence evaluated at 3.5 and 4.5 years of age (all antigens)			NA	NA	436 at 4.5y (167 Hexaxim without HB at birth; 102 Hexaxim with HB at birth)	Hexaxim induces good persistence of antibodies to each antigen up to pre-school age in children given Hexaxim at 6-10-14 weeks & booster (most challenging schedule)	[35]

(*Continued*)

Table 2. (Continued).

Trial	Country	Year	HB vaccine at birth	PS B Schedule	Control vaccine(s)	Co-administration with other vaccines	Total number of subjects (Subjects receiving Hexaxim)	Key findings	Reference
A3L28 (Follow-up of A3L24/ A3L27)	Colombia	2013-2015	Yes	2-4-6 months & booster; persistence evaluated at 3.5 and 4.5 years of age (all antigens)	NA	NA	538 at 4.5 y (213 Hexaxim as PS & B, 200 Hexaxim as PS, 125 Hexaxim as B)	Hexaxim induces good persistence of antibodies to each antigen up to pre-school age in children given Hexaxim at 2-4-6 months & booster	[35]
A3L47†/ A3L49 (Follow-up of A3L12)	Thailand	2016	Yes	2-4-6 months (No booster); persistence evaluated at 9-10 years of age (HB)	NA	NA	150 (71)	The kinetics of long-term anti-HBs antibody persistence were similar following a primary series of Hexaxim or Infanrix hexa The response to a subsequent HB challenge re-vaccination was strong and similar in each group, demonstrating persisting immune memory	[36]
A3L00052 (Follow-up of A3L38)	Finland Sweden	2019	No	3-5 months & booster (11-12 months); persistence evaluated at 6 years of age (HB)	NA	NA	189 (91)	At 6 years, 53.8% of children who received Hexaxim and 73.5% of children who received Infanrix hexa had seroprotective anti-HBs antibodies (≥10 mIU/mL), increasing to 96.7% and 95.9% respectively postchallenge, confirming a strong anamnestic response in primed vaccinees	[37]

This table includes trials conducted in healthy infants/toddlers; trials conducted in specific populations (A3L44 [46], A3L53 [44]) are not shown in this table.
B, booster; EPI, Expanded Program on Immunization; HB, hepatitis B; Men, meningococcal; MMR, measles-mumps-rubella; NA, not applicable; PCV, pneumococcal conjugate vaccine; PS, primary series.
†This was a laboratory analysis on retention sera from another clinical trial (WHO UTN: U1111–1161–2649) which enrolled A3L12 cohort subjects and so was not a clinical trial per se (no investigational product was used).

rotavirus (RV) vaccines [21,23,31,33], meningococcal C (MenC) [23] and MenACWY conjugate vaccines [24]. Additionally, a clinical trial to evaluate the co-administration of Hexaxim with MenB vaccine (Bexsero [GSK]) is ongoing (EudraCT Number: 2019–002585–12).

2.1.1.6. *Mixed hexavalent-pentavalent-hexavalent primary series schedule at 2, 4, 6 months of age.*

A mixed hexavalent-pentavalent-hexavalent primary vaccination schedule after a birth dose of standalone HB vaccine was evaluated in one post-licensure trial in Spain [34] (Table 2). The primary series immune response for each antigen was strong (seroconversion rates [PT, FHA] >89% and seroprotection rates [other antigens] >99%), and a Pentaxim booster was immunogenic, with both primary series and booster responses aligned with a hexavalent primary series followed by a pentavalent or hexavalent booster vaccination [16,31]. These data support the use of Hexaxim in a hexavalent-pentavalent-hexavalent primary series after a birth dose of HB vaccine.

2.1.1.7. *Persistence of immunity.*

Antibody persistence was evaluated for all antigens up to school age (3.5 and 4.5 years of age) in South Africa and Colombia [35], and longer term for HB in Thailand (9–10 years of age) [36] and Finland (6 years of age) [37].

In the South Africa and Colombia follow-ups [35], Hexaxim induced good antibody persistence up to 4.5 years of age for each antigen, notably following the challenging 6, 10, 14 week primary vaccination schedule in South Africa, where persistence of anti-HBs antibodies (≥10 mIU/mL) was observed in 73% (without HB vaccine at birth) to 96% (with HB vaccine at birth) of individuals and persistence of seroprotective levels of anti-D, anti-T, anti-polio 1, 2, 3 and anti-PRP in ≥97% of children. For pertussis, anti-PT and anti-FHA GMCs were generally similar between Hexaxim and control vaccine(s).

The trial in Thailand provides a unique opportunity to evaluate anti-HBs antibodies following a 2, 4, 6 month schedule after a birth dose of HB vaccine and with no HB booster vaccine (according to the national vaccination calendar in Thailand) (Figure 4). Anti-HB antibodies decreased to a similar extent at 12–18 months of age following Hexaxim (90.8% ≥10 mIU/mL) or Infanrix hexa (96.5% ≥10 mIU/mL) and continued to decline to a similar extent by 9–10 years of age (49.3% and 42.9% for Hexaxim and Infanrix hexa, respectively). Following receipt of an HB challenge dose at 9–10 years of age (ie, a HB re-vaccination given as part of a clinical study to evaluate response to HB exposure), anti-HBs antibody levels increased in each group, with seroprotection rates of 92.8% and 98.7% for Hexaxim and Infanrix hexa, respectively. These data confirm that long-term protection against HB appears to be more dependent on persistence of immune memory (memory T and B cells) rather than persistence of high levels of antibody. This is further supported by data from the Finland trial, in which HB seroprotection rate after a 3, 5, and 11 month vaccination schedule was lower for Hexaxim than Infanrix hexa at 6 years of age (53.8% versus 73.5% ≥10 mIU/mL), but increased to a similar level following an HB challenge vaccination (96.7% and 95.9%, respectively), confirming persistent immune memory.

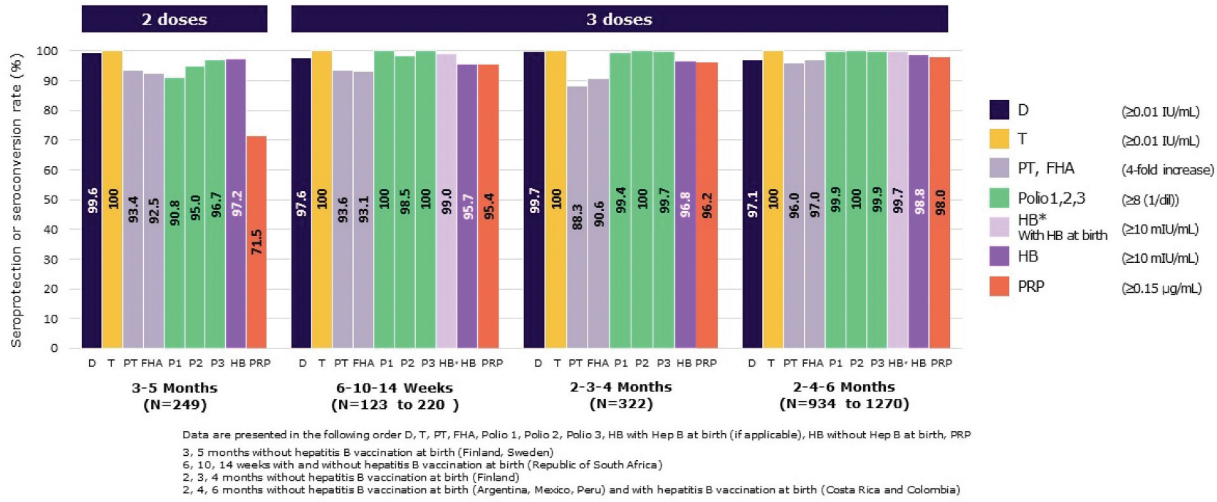


Figure 2. Seroprotection/Seroconversion rates 1 month after a primary vaccination with 2 or 3 doses of Hexaxim.

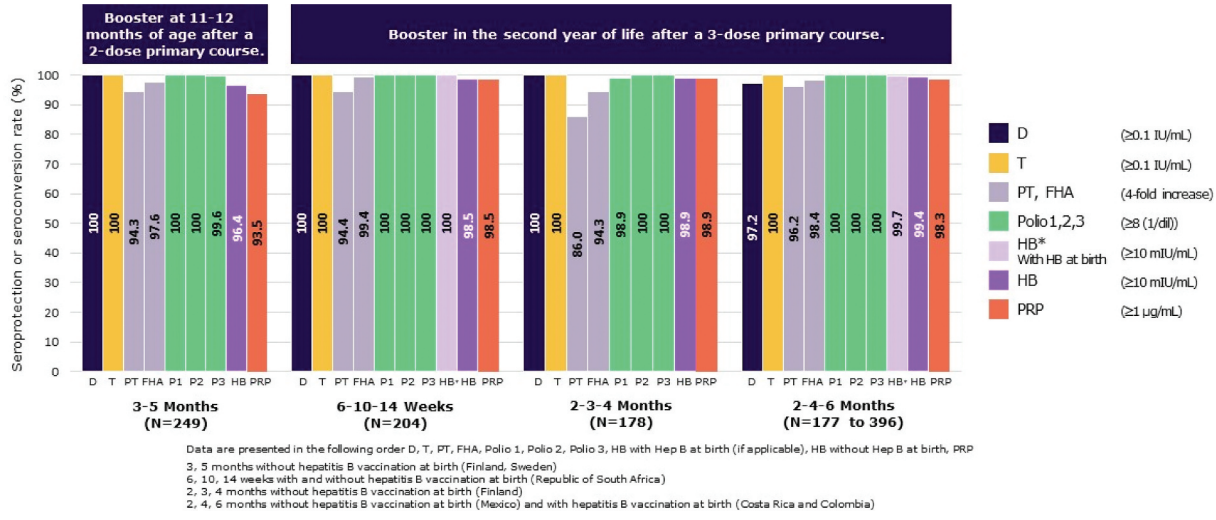


Figure 3. Seroprotection/Seroconversion rates 1 month after a booster vaccination of Hexaxim in the second year of life.

2.1.2. Safety

A large-scale safety trial was conducted in approximately 2000 subjects (of whom 1422 subjects received Hexaxim) in Mexico and Peru [25]. In this RCT, a whole-cell pertussis (wP)-containing, reconstituted pentavalent vaccine (DTwP-HB/Hib) was used as a comparator in conjunction with OPV. Hexaxim and the comparator vaccines were administered in a 2, 4, 6 month primary series schedule. This trial achieved its primary objective of demonstrating non-superiority of the incidence of severe fever (≥39.6°C) following Hexaxim administration versus the comparator vaccines (4.0% [95% CI 3.0–5.1] versus 5.5% [95% CI 4.0–7.5], respectively, after any dose); however, the study cannot conclude that Hexaxim induced a lower incidence of high fever than the comparator.

Additionally, the incidence of solicited reactions, which was evaluated in the 7 days post-vaccination, was consistently higher for the comparator vaccines than for Hexaxim (eg, 92.7% versus 74.8% of participants reported fever (≥38.0°C)). The same trend was observed for severe injection site and systemic reactions. These data are well aligned with the better safety profile of aP-containing vaccines versus wP-containing vaccines, which is well documented [38]. A single hypotonic hyporesponsive episode (HHE) occurred in one subject approximately 7 hours after the first dose of Hexaxim, although the occurrence of HHE in one of the two groups was not unexpected based on the documented incidence following vaccination [39–41], given the large overall sample size and the number of doses administered in the trial (approximately 6000 doses of either Hexaxim or the comparator vaccines). No other episodes of HHE have been observed in any other Hexaxim trial.

The trial in South Africa (in which 380 subjects out of 622 received Hexaxim) also included a wP-containing comparator (DTwP/Hib) plus HB and OPV vaccines [17,18]. In this trial, following a 6, 10, 14 week primary series and a booster at 15–18 months of age, the incidence of solicited injection site reactions was consistently lower in subjects who received Hexaxim compared to the wP-containing vaccine, although solicited systemic reactions occurred to a similar extent in each group. It should be noted

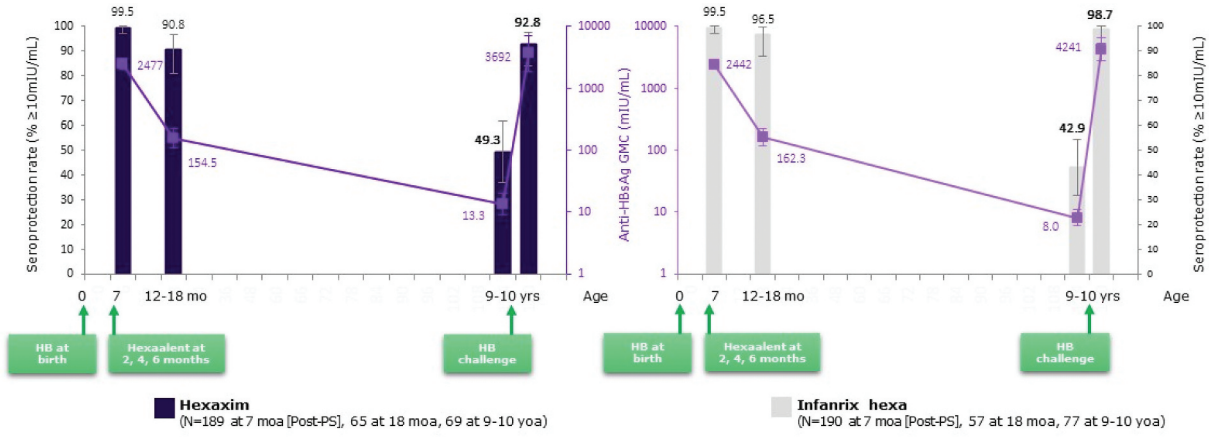


Figure 4. Persistent immune memory to HB following 2, 4, 6 month primary series vaccination of Hexaxim and no booster.
Source: Kosalaraska et al (2018) [36].

that this trial was not powered to evaluate differences in safety profile between the aP- and wP-containing vaccines, but overall the results are considered to be aligned with the large-scale safety trial described above [25].

All other comparator vaccines used in Hexaxim trials contained aP rather than wP components (Pentaxim and monovalent HB vaccine or Infanrix hexa) (see Table 2). In all these trials the safety profile of Hexaxim was aligned with that of the aP-containing comparator vaccine.

Results from an integrated analysis of safety from 6 RCTs in which Infanrix hexa was used as comparator [21,28–31,33] are provided in Table 3. For primary series vaccination, all solicited injection site and systemic reactions (any grade and Grade 3) were reported to a similar extent for both vaccines except for injection site pain (any grade, Grade 3) and irritability (Grade 3), which were reported slightly more frequently for Hexaxim based on non-overlapping 95% confidence intervals. It was noted that fever Grade 3 (≥39.6°C) was reported with a similar frequency for Hexaxim (1.2% [95%CI: 0.9; 1.7]) and Infanrix hexa (1.3% [95%CI: 0.8; 2.1]). Other authors, using a different methodology (aggregate data meta-analysis), concluded that odds ratios of analyzed local and systemic solicited adverse reactions after primary vaccination with Infanrix hexa appeared to be slightly lower than with Hexaxim [42].

Two trials compared a full primary series and booster schedule of Hexaxim to Infanrix hexa [21,33] and reported similar overall incidences of solicited reactions for both vaccines.

Overall, there were no clinically meaningful differences in safety profiles between Hexaxim and Infanrix hexa.

2.2. Hexaxim in specific populations

2.2.1. Preterm infants

Since licensure there has been no contraindication to use Hexaxim in preterm infants, with many countries recommending the same primary vaccination schedule in preterm (according to chronological age) and term infants. A post-marketing surveillance study in Italy reported a good safety profile in 339 preterm infants, including 13% ($n = 44$) born between the 28th–32nd week of gestation (very preterm) and 0.3% ($n = 1$) born before the 28th week of gestation (extremely preterm) [43]. After Hexaxim vaccination, injection site pain was the most common solicited reaction (35.7% of infants) with erythema, swelling, and induration also reported in approximately 25% of infants. These findings are in line with the safety profile in term infants.

A prospective, observational study included a cohort of preterm infants ($N = 105$) who received Hexaxim at 8, 12, 16 weeks of age (primary series) and at 13 months of age (booster) in Belgium [44], as well as a comparator group of term infants ($N = 127$). In this study, the immune response to primary vaccination was similar in preterm and term infants for all antigens except HB and PRP, for which the response was similar in term and preterm infants post-booster, suggesting adequate priming after the 3-dose primary schedule. These data are similar to findings from a review of Infanrix hexa in preterm infants [45], and support the use of Hexaxim as primary and booster vaccination in preterm infants.

A study assessing the use of Hexaxim in preterm infants born between 24 and 36 weeks gestation is currently ongoing in Spain (EudraCT number: 2018–004581–34).

2.2.2. Infants of women vaccinated with Tdap in pregnancy

An evaluation of the immune response to primary series and booster vaccination in infants born either to women vaccinated with a Tdap vaccine in pregnancy ($N = 199$) or to women not vaccinated with Tdap in the previous 5 years ($N = 33$) was included in the prospective, observational study in Belgium [44]. As expected, infants of Tdap-vaccinated women had higher pre-vaccination levels of anti-D, anti-T, anti-PT, and anti-FHA antibodies. There were generally only small differences in seroprotection rates (anti-D, anti-T, anti-polio 1, 2, 3, anti-HBs, and anti-PRP) or the percentage of infants with anti-PT/anti-FHA antibodies ≥8 EU/mL after both the primary series and booster vaccination. These data supported the use of Hexaxim in infants born to Tdap-vaccinated or Tdap-unvaccinated women.

Table 3. Solicited reactions after any infant vaccination – integrated analysis of Hexaxim versus Infanrix hexa.

	Hexaxim† (*N* = 3191)		Infanrix hexa† (*N* = 1385)	
	%	(95% CI)	%	(95% CI)
Subjects experiencing at least one:				
Injection site reaction	84.6	(83.3; 85.8)	78.0	(75.7; 80.2)
Injection site pain (any grade)	77.6	(76.1; 79.0)	68.0	(65.5; 70.5)
Grade 3	10.3	(9.2; 11.4)	6.1	(4.9; 7.5)
Injection site erythema (any grade)	49.6	(47.8; 51.3)	45.2	(42.5; 47.8)
Grade 3	1.6	(1.2; 2.1)	1.2	(0,7; 2.0)
Injection site swelling (any grade)	34.8	(33.1; 36.5)	32.5	(30.0; 35.0)
Grade 3	1.5	(1.1; 2.0)	0.9	(1.5; 1.6)
Systemic reaction	93.7	(92.8; 94.6)	93.2	(91.8; 94.5)
Pyrexia (any grade)	40.7	(39.0; 42.4)	41.7	(39.0; 44.3)
Grade 3	1.2	(0,9; 1,7)	1.3	(0,8; 2,1)
Vomiting (any grade)	34.9	(33.2; 36.6)	33.6	(31.2; 36.2)
Grade 3	1.5	(1.1; 2.0)	1.2	(0,7; 2.0)
Crying (any grade)	73.7	(72.2; 75.3)	71.4	(69.0; 73.8)
Grade 3	6.3	(5,4; 7,2)	4.4	(3,4; 5,6)
Somnolence (any grade)	58.3	(56.5; 60.0)	61.0	(58.4; 63.6)
Grade 3	4.2	(3,5; 5.0)	2.5	(1.8; 3.5)
Decreased appetite (any grade)	45.8	(44.1; 47.6)	41.7	(39.0; 44.3)
Grade 3	2.4	(1.9; 3.0)	1.6	(1.0; 2.4)
Irritability (any grade)	81.0	(79.6; 82.4)	78.7	(76.4; 80.8)
Grade 3	8.5	(7.6; 9.5)	5.9	(4.7; 7.3)

†Compilation from trials A3L11, A3L12, A3L17, A3L24, A3L38, A3L39.

Grades 1, 2 and 3 pain were defined as 'minor reaction when injection site is touched,' 'cries or protests when injection site is touched,' and 'cries when injected limb is moved or the movement of the injected limb is reduced.' For erythema and swelling, a diameter of <2.5 cm was assessed as Grade 1, from 2.5 to <5 cm as Grade 2 and ≥5 cm as Grade 3. Grades 1, 2 and 3 pyrexia were defined as temperature ≥ 38–≤38.5°C, >38.5 to ≤ 39.5°C and > 39.5°C, respectively. Other systemic symptoms were defined as vomiting (Grade 1, 1 episode/day; Grade 2, 2 to 5 episodes/day; Grade 3, ≥6 episodes/day or requiring parenteral hydration), crying (Grade 1, <1 hour; Grade 2, 1–3 hours; Grade 3, >3 hours), somnolence (Grade 1, unusually sleepy; Grade 2, not interested in surroundings or did not wake up for a meal; Grade 3, sleepy most of the time or difficult to wake up), decreased appetite (Grade 1, eating less than normal; Gade 2, missed 1 to 2 meals; Grade 3, missed ≥ 3 meals), and irritability (Grade 1, easily consolable; Grade 2, requiring increased attention; Grade 3, inconsolable).

2.2.3. Immunocompromised infants

A trial was conducted in South Africa in HIV-infected (*N* = 14) and HIV-exposed but uninfected infants (*N* = 50) who received a 6, 10, 14 week primary series and a booster at 15–18 months of age [46]. It was difficult to recruit infected infants due to widespread pre- and peri-natal retroviral treatment, but despite the small number of infants, primary series and booster immune responses were strong except for anti-HBs ≥10 mIU/mL, which was lower in infected than uninfected infants (78.6% post-primary series and 75.0% post-booster in the HIV-infected cohort compared to 100% post-primary series and post-booster in the uninfected cohort). There were no safety concerns in either group, with no clinically meaningful differences between groups. The immune responses as well as the incidences of solicited AEs were broadly comparable with data from previous trials in healthy infants, notably the one conducted in South Africa in a similar population in terms of ethnicity and vaccination schedule [17,18].

2.3. Post-marketing surveillance

2.3.1. Overview of spontaneous adverse events reported post-licensure

The review and analyses of spontaneous data extracted from the Sanofi global safety database from 1 June 2013 to 17 April 2023 identified 18,320 spontaneous AE reports worldwide following Hexaxim vaccination, with more than 180 million doses of Hexaxim having been distributed. According to the Guideline on Good Pharmacovigilance Practices (GVP) Module VI, all these spontaneous reports are considered as related to Hexaxim vaccination by default [47]. Of these 14,958 (81.6%) were non-serious and 3,362 (18.4%) were serious. The five countries with the highest incidences of AE reports were: Italy (7,611 reports [41.5%]), Malaysia (2,513 reports [13.7%], Germany (1,206 reports [6.6%], South Africa (1,039 reports [5.7%]), and France (759 reports [4.1%]) (Table 4).

The ten most frequently reported events (non-serious and serious) were: fever, injection site erythema, crying, injection site swelling, rash, irritability, injection site pain, diarrhea, vomiting, and decreased appetite, with a frequency ranging between 0.36 and 5.14 per 100,000 doses distributed (Table 4). Of the serious reported events, the 5 most frequent were: fever, convulsion with or without fever, crying, hypotonic-hyporesponsive episode/hypotonia, and vomiting.

These findings are consistent with the known safety profile of Hexaxim.

2.3.2. Adverse events of special interest

Extensive limb swelling, anaphylactic reaction, convulsion with or without fever, and HHE have been closely monitored since the launch of Hexaxim. For each of these adverse events of special interest, reporting rates were calculated using a conservative approach by including all cases whether clinically confirmed or not. The respective reporting rates have been stable since 2017. Reporting rates for anaphylactic reaction range from 0.08 to 0.11 cases per 100,000 doses

Table 4. Characteristics of AE reports and the 10 most common events reported after DTaP-IPV-HB-Hib vaccination (June 1, 2013 to April 17, 2023).

	Number of AE reports	%		Number of AE reports	%
Seriousness			***Countries***		
All ages at vaccination (N = 18,320)			*All ages at vaccination (N = 18,320)*		
Non serious	14958	81.6	Italy	7611	41.5
Serious	3362	18.4	Malaysia	2513	13.7
2–3 years old at vaccination (N = 210)			Germany	1206	6.6
Non serious	154	73.3	South Africa	1039	5.7
Serious	56	26.7	France	759	4.1
3–6 years old at vaccination (N = 221)			Other countries	5192	28.3
Non serious	175	79.2			
Serious	46	20.8			

Event‡	MedDRA prefered term	N† (%)	RR per 100 000 doses distributed
Fever	Pyrexia/Body temperature increased/Hyperpyrexia/Hyperthermia	9295 (22.5)	5.14
Injection site erythema	Vaccination site erythema/Injection site erythema/Administration site erythema/Application site erythema/Puncture site erythema	1762 (4.3)	0.97
Crying	Crying	1484 (3.6)	0.82
Injection site swelling	Vaccination site swelling/Injection site swelling/Application site swelling/Administration site swelling/Puncture site swelling	1480 (3.6)	0.82
Rash	Rash/Rash maculo-papular/Rash erythematous/Rash macular/Rash pruritic/Rash papular	1222 (3.0)	0.68
Irritability	Irritability/Irritability post-vaccinal	997 (2.4)	0.55
Injection site pain	Vaccination site pain/Injection site pain/Administration site pain/Application site pain/Puncture site pain	940 (2.3)	0.52
Diarrhea	Diarrhoea	937 (2.3)	0.52
Vomiting	Vomiting	701 (1.7)	0.39
Decreased appetite	Decreased appetite	642 (1.6)	0.36

‡Includes non-serious and serious events.
†Percentage calculated from total number of reported events (*N* = 41,378); Cumulative number of doses distributed as per April 30, 2023 (*N* = 180,791,895).
MedDRA, Medical Dictionary of Regulatory Activities; RR, Reporting Rate.

distributed, for extensive limb swelling from 0.16 to 0.3, for convulsion with or without fever from 0.27 to 0.33, and for HHE from 0.08 to 0.14 (Figure 5).

2.3.3. Sudden Infant Death Syndrome/Sudden Death

Sudden infant death syndrome (SIDS) is defined as death that occurs in the first year of life and remains unexplained after autopsy [48]. Preferred terms of SIDS and Sudden Death were used to search the Sanofi safety database to identify all cases of Sudden Death.

Cumulatively, 23 cases of SIDS/Sudden Death were reported in infants in their first year of life following Hexaxim vaccination, mostly after the first dose. Of these, 20 cases had a time to onset ≤15 days and occurred in the European Economic Area. Observed to Expected (O/E) analyses stratified on time to onset were performed [49], and sensitivity analyses were also conducted to test assumptions on the number of doses administered and under-reporting. The O/E analysis did not show any ratio increase for SIDS/Sudden Death in European countries irrespective of the risk window and assumptions made (number of doses administered and the reporting rate) (Table 5). The O/E ratios (95% confidence interval) for a risk window of 0–3, 0–15, and 0–63 days were 0.34 (0.19–0.60), 0.12 (0.07–0.19), and 0.04 (0.02–0.05) respectively. These results show that the number of deaths reported did not exceed the number expected to occur by chance.

2.3.4. Use in children >2 years of age up to 6 years of age

Of all AE reports, 431 (2.3%) were from children aged 2 to 6 years (210 reports in children aged 2–3 years and 221 reports in children aged >3–6 years) who had been exposed to Hexaxim. Most cases (329/431 [76.3%]) were non-serious (Table 4). The most frequently reported events were fever and injection site reactions, consistent with those reported in infants <2 years old.

2.4. Efficacy/Effectiveness against pertussis

Historical efficacy trials confirmed that registered aP vaccines are immunogenic and effective in preventing pertussis, showing consistent, high efficacy ranging from 71% to 93% [50].

The efficacy of the aP antigens contained in Tetraxim, Pentaxim, and Hexaxim (PT and FHA) was demonstrated in a double-blind RCT (the Senegal efficacy trial, 1990–94) for the DTaP combination vaccine (DTaP backbone) [51]. Vaccine efficacy (VE) against WHO-defined classical pertussis (≥21 days of paroxysmal cough) was 85% (95% CI: 66, 93) for the aP antigens contained in Tetraxim, Pentaxim, and Hexaxim and 96% (95% CI: 86, 99) for the DTwP vaccine (median follow-up period of 1.58 years, ranging from 0.01 to 4.25 years) [51]. Since that time, real-world evidence from national surveillance programs has demonstrated the effectiveness of aP vaccines in protecting against pertussis.

Surveillance data from Sweden (1996–2022) demonstrated that the incidence of pertussis among infants decreased after 1996 when aP was introduced into the National Immunization Program [1,52]. In a recent evaluation of the real-world effectiveness of aP vaccines in Sweden, the various aP vaccines (including the aP antigens used in Pentaxim and Hexaxim) used since 1996 continued to show high effectiveness, sustained at 95–97% between 2007 and 2018 [52,53].

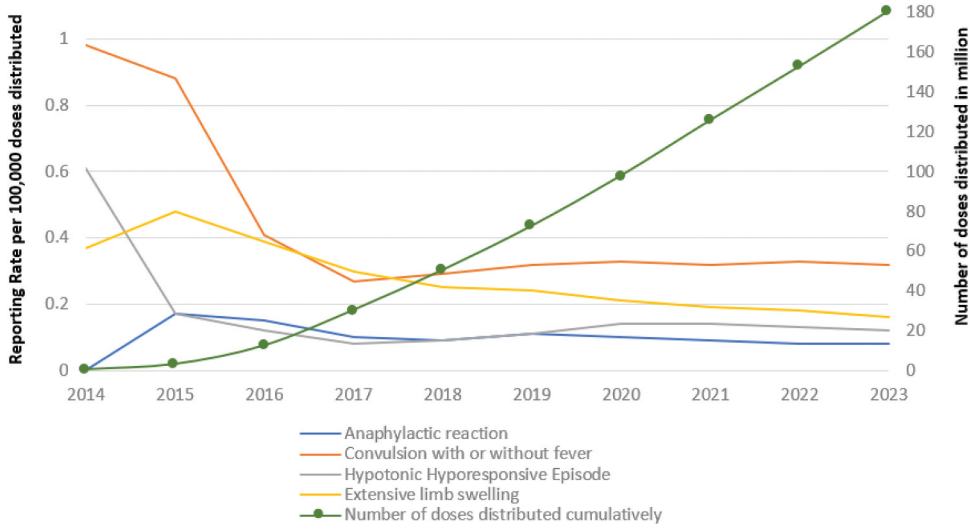


Figure 5. Evolution of the reporting rate for adverse events of special interest.

Table 5. Observed versus expected analysis for Sudden Infant Death Syndrome/Sudden Death in the European Economic Area.

Assumptions (Dose administered/ Reporting Rate)	Time to onset (days)	Number of observed cases	Number of expected cases	O/E ratio (95% CI)
100/100	4	12	35	0.34 (0.19–0.60)
100/100	16	17	142	0.12 (0.07–0.19)
100/100	64	20	567	0.04 (0.02–0.05)
75/75	4	15	27	0.56 (0.34–0.94)
75/75	16	21	106	0.20 (0.13–0.31)
75/75	64	25	425	0.06 (0.04–0.09)

The observed versus expected analysis were conducted using methods described by Mahaux et al.

Twenty-three cases were retrieved from Sanofi safety database using the MedDRA (Version 24.1) PTs 'Sudden Death' and 'Sudden Infant Death Syndrome.' Twenty cases were from the EEA and classified based on their time to onset following vaccination with DTaP-IPV-HB-Hib (0–3 days, 0–15 days, and 0–63 days). One case had unknown time to onset and has been included in the three risk windows for conservative estimates. All reported cases were included in the O/E analyses whatever their level of diagnostic certainty (Brighton Collaboration assessment).

The background incidence rate of SIDS was obtained from countries reporting SIDS in Eurostat database (ICD code R-95 'Sudden Infant Death Syndrome'). The rates ranged from 0.1 to 0.5 per 1000 live births per year [100]. As a conservative approach the lowest rate was selected.

Sensitivity analyses were conducted assuming 1) only 75% of the doses distributed have been administered and 2) an under-reporting of 25% (only 75% of the cases have been reported and included in Sanofi safety database) [101].

Similarly, in Mexico where a 2 component aP vaccine (Pentaxim then Hexaxim) has been used exclusively since 2008, surveillance data covering the 11-year period post introduction (2008–2019) demonstrated that the aP antigens used in Pentaxim, and Hexaxim effectively protect infants against confirmed pertussis [54,55]. From 2011 to 2014, among 192 children <1 year of age who were hospitalized with confirmed pertussis in Mexico, ≥2 doses of aP vaccine were associated with a reduction in the number of pertussis cases, and no case of pertussis was detected among infants who had received three doses [54]. Furthermore, analyses of the nationwide data on pertussis surveillance in Mexico (2008–2019) demonstrated that three doses of Pentaxim or Hexaxim elicited 97.2% protection against laboratory-confirmed pertussis, a value comparable to that observed for the wP vaccine (96.4%) [55].

A recent hospital-based surveillance study from Malaysia, where a 2 component aP vaccine (Pentaxim and Hexaxim) has been used exclusively since 2008, showed that the majority of confirmed pertussis cases (89.3%) occurred in infants too young to be fully vaccinated or under-vaccinated for their age [56].

Protection against pertussis, whether resulting from infection or vaccination, wanes over time. Most available real-world evidence from disease surveillance and effectiveness studies suggests that regardless of the vaccine type, protection elicited by a primary series and toddler booster schedule wanes by school-age, requiring regular booster to extend pertussis protection.

2.5. Vaccination strategies and programmatic benefits of Hexaxim

2.5.1. Transition from wP-containing to aP-containing vaccines

Some authors have suggested that the transition from wP- to aP-containing vaccines could be one of the causes of a resurgence in pertussis incidence that has been observed in some countries in recent years, including Australia and the U.S.A. Routine infant and toddler vaccination is associated with waning antibody levels over time, which can lead to increased incidences of vaccine-preventable diseases, including pertussis, and it has been postulated that pertussis resurgence could be due to more rapidly waning immunity

following aP vaccination [50,57–61]. However, the causes of pertussis resurgence are complex and multi-factorial, including other aspects such as genetic modifications of *B. pertussis* or reduced vaccine coverage linked to increased vaccine hesitancy, or asymptomatic transmission including among adolescents and adults, or even the possibility of increased pertussis reporting being an artifact due to improved diagnostic methods [50]. Pertussis resurgence has been observed in countries using wP vaccines [62] and in some countries pertussis resurgence started prior to the introduction of aP-containing vaccines [54,63], and despite fluctuations in pertussis incidence over time, VE against pertussis, as described earlier, has been shown to be comparable in Mexico for the periods of vaccination using wP-containing (2000 to 2007) and aP-containing (2008 to 2019) vaccines.

Experience in other countries supports the successful transition from wP to aP vaccination. In Sweden, aP-containing vaccines (either Pentaxim or Infanrix-IPV-Hib) were introduced in 1997 in a 3, 5, 12 month of age schedule, following withdrawal of the wP vaccine in 1979. Vaccination coverage rapidly reached 98–99% (within 2–3 months), and by 3 years following the introduction of aP vaccines pertussis incidence had dropped by 80–90% and was comparable to historic incidences during the earlier period of wP vaccination [53,64–66]. In other countries, including Germany, Japan, and the U.S.A., the introduction of aP-containing vaccines has also been shown to be effective in reducing the incidence of pertussis disease [67–69].

These data taken together show that aP-containing vaccines provide high levels of protection against pertussis disease in infants and toddlers, and that the transition from wP- to aP-containing vaccines, including Pentaxim and Hexaxim, has been successful in the continuous control of pertussis control in a range of countries and vaccination schedules. Additionally, transitioning to aP-containing vaccines is likely to improve parental satisfaction and acceptability of vaccination; in Chile, following a switch from DTwP-HepB/Hib + oral poliovirus vaccine (OPV) to Hexaxim in 2018, more parents reported no changes in their daily life and no disruption in other children's sleep [70]. Lastly, transitioning to aP-containing vaccines may bring economic benefits linked to reducing AE incidence and reduced logistic and social costs [71–73].

2.5.2. Contribution to polio eradication

The transition from OPV to IPV is crucial in the global effort to eradicate polio, and planning for OPV cessation is part of the Global Polio Eradication Initiative (GPEI) Strategy 2022–2026 [74]. The use of OPV (inducing a strong mucosal immune response at the gastrointestinal level) has been fundamental to reducing the global incidence of polio by controlling widespread circulation of polioviruses. However, Vaccine Associated Paralytic Polio (VAPP) and Vaccine Derived Polioviruses (VDPVs) are rare but serious risks associated with the continued use of OPV, and since 2016 more than 40 countries have been affected by outbreaks caused by VDPVs [75].

Inactivated poliovirus vaccine (inducing a strong humoral immune response) is highly effective in preventing paralytic disease caused by all three types of polioviruses and has several efficacy and safety advantages over OPV, including no risk of VAPP or VDPV [76]. Since 2022, the WHO recommends that all children worldwide should be fully vaccinated against polio with 3–5 doses of polio vaccine, including at least two doses of IPV [4]. Non-inferiority of the immune response to IPV following Hexaxim infant primary series and toddler booster vaccination compared to OPV has been demonstrated [17,18]. With the inclusion of the trivalent IPV antigen into its fully liquid presentation that promotes improved compliance to pediatric vaccination schedules, Hexaxim is well placed to support the GPEI-mandated transition to IPV in the endgame strategy for the global eradication of polio.

As the safest polio vaccine available, IPV is a cornerstone in the endgame of global polio eradication to eliminate all types of paralytic poliomyelitis, including vaccine-derived polio [77]. In countries with low circulation of polio, a full IPV regimen may be a better option than a mixed OPV and IPV regimen, and Hexaxim may be a valuable option in this context, with many developing countries moving to a full IPV schedule recently following recommendations from medical societies [78,79].

2.5.3. Transition from pentavalent to hexavalent vaccines

The WHO recommends that all infants (including low birth weight and premature infants) should receive their first dose of HB vaccine as soon as possible after birth, ideally within 24 hours, followed by 2 or 3 additional doses to complete the primary series [6]. For a 3-dose HB vaccination schedule, the birth dose of monovalent HB vaccine can be followed by second and third doses of either monovalent vaccine or as part of a combined vaccine, given at the same time as the first and third doses of DTP-containing vaccine. Where 4 doses of HB vaccine are given, the additional dose of HB vaccine given as part of the second dose of routine DTP-containing vaccine does not cause any harm. Several Hexaxim studies have included a birth dose of HB vaccine [17,19,22,29,31,32,34,80] (see Table 2).

The transition from pentavalent vaccination and monovalent HB vaccine to hexavalent vaccination means that the second dose of HB vaccine would be administered at 2 months of age rather than 1 month of age. A recent systematic review of HB infant vaccination programs in South East Asia and Western Pacific regions [81] reported a study where no evidence of any higher risk of immunoprophylaxis failure was shown when HB vaccine is administered in a 0, 2 month regimen compared to 0, 1 months (odds ratio: 0.77) [82]; furthermore, the same review [81] reported that HBV surveillance in Australia has shown a continuous decline of HB in children since the introduction of a 0, 2, 4, 6 month schedule compared to the previous 0, 1, 6 month schedule [83]. A potential option for maintaining three doses of HB vaccine could be a birth dose of HB vaccine followed by a mixed hexavalent-pentavalent-hexavalent schedule, as described earlier and evaluated in a trial in Spain [34], which showed >99% anti-HB seroprotection rate following HB vaccination at 0, 2, 6 months of age. The switch from pentavalent to hexavalent vaccination has been made in Malaysia and Singapore [84].

The benefits of a ready-to-use vaccine make the transition from pentavalent to hexavalent vaccination advantageous in terms of both healthcare professional and parental satisfaction [70,85], supporting the widespread implementation of Hexaxim vaccination of infants and toddlers.

2.6. Manufacturing and benefits of a ready-to-use vaccine

2.6.1. Manufacturing aspects

Unlike drugs that contain small chemical entities, vaccines contain antigens that are large complex molecules (eg, protein, polysaccharide) obtained from bioproduction. Thus, the manufacturing of vaccines is complex and vaccines are characterized less by their chemical composition than by their manufacturing process, which should be fully controlled [86,87]. Quality controls performed throughout the manufacturing process are key to demonstrate the robustness and reproducibility of the manufacturing process.

For multivalent vaccines such as Hexaxim, with nine different antigens that are diverse in their nature as well as the inclusion of an adjuvant (see Table 1), manufacturing is particularly challenging [87]. The pertussis antigens (PT and FHA) obtained from the same fermentation process of *B. pertussis* are separately purified using highly selective methods [88] resulting in low levels of residual pertussis endotoxins that can contribute to reactogenicity in wP vaccines [89]. When all active ingredients are available, the fully liquid hexavalent vaccine is formulated by sequential addition of the individual antigens according to precise operational protocols to achieve a homogeneous and consistent formulation [88], allowing the correct adsorption level onto the aluminum gel for some antigens (eg, HBsAg) and limiting interactions for others (eg, PRP~T) [87,90,91]. Robust quality control is therefore crucial, and in the manufacturing process for Hexaxim this accounts for 70% of the total cycle time [87] and includes more than 1200 individual quality tests (Sanofi, data on file). Quality control tests are performed throughout the manufacturing cycle, including controls for raw materials, for each antigen, for the final bulk product, and for filled product. Quality control tests are performed to check the safety (eg, sterility, control of inactivation of pertussis toxin), quality (eg, general parameters such as pH, osmolality; identity and purity of each antigen) and potency (eg, serological activity) of the vaccine [87,88]. Finally, before being shipped to health care professionals, vaccines are controlled and released by both the manufacturer and an external National Control Laboratory.

The rigorous manufacturing of Hexaxim is key for the consistency of vaccine lots, which was demonstrated in lot-to-lot consistency trials in Mexico [28] and Colombia/Costa Rica [31]. Such consistency translates into predictable and reliable clinical performance (ie, safety, efficacy) between lots over time and up to the end of the vaccine's shelf-life (ie, 48 months).

2.6.2. Benefits of a ready-to-use vaccine

The fully liquid presentation of Hexaxim, not requiring reconstitution of the Hib component prior to administration (as this is the case for another hexavalent vaccine), helps to simplify healthcare practice by eliminating time needed for reconstitution and reducing the risk of immunization errors [92].

As well as missed reconstitution of the Hib component, vaccine handling errors include inadequate shaking of vaccines, incomplete aspiration of reconstitution vials, and spillage or leakage during reconstitution, and have been reported by 20–30% of physicians and 6–7% of nurses surveyed in South Korea [93]. Healthcare providers from different countries are concerned about the risks of errors during vaccination, acknowledge that mistakes can be made when reconstituting vaccines, and have shown a preference for fully liquid presentation [92,94–96]. Furthermore, time efficiencies associated with the use of fully liquid vaccines can contribute to reducing workload in healthcare centers and allow potential cost savings [85,94,95,97].

3. Discussion

Combination vaccines, including Hexaxim, are key to many infant immunization programs, contributing to improved childhood health worldwide. They offer several advantages over administering individual vaccines separately. First, combination vaccines reduce the number of injections required, minimizing discomfort and distress for both infants and parents. Second, they improve vaccine compliance and coverage rates, through the administration of multiple antigens simultaneously. Third, combination vaccines contribute to cost-effectiveness and operational efficiency by reducing healthcare resources and logistical challenges associated with multiple vaccine administration visits [98].

Hexaxim is an established, fully liquid, ready-to-use, hexavalent vaccine for primary and booster vaccination of infants developed for use worldwide outside the U.S.A. It is built on established D, T, aP, IPV, and PRP~T antigens, includes an HB antigen specifically developed for inclusion into Hexaxim, and is preservative-free. This hexavalent vaccine was widely assessed in clinical trials prior to its first licensure, and clinical evaluation has continued for the 10 years since licensure, representing ongoing commitment to expand the benefits of Hexaxim to new populations. Overall, more than 25 clinical trials have been completed, involving more than 7200 infants/toddlers. These trials have been published routinely, providing an extensive body of clinical data pertaining to Hexaxim that are available in the scientific literature.

The clinical data have consistently shown non-inferior immunogenicity to all comparator vaccines and have confirmed the good safety profile of Hexaxim when administered as a primary series and booster vaccination in a 3 + 1 or 2 + 1 schedule. Hexaxim has been shown to be highly immunogenic in a wide range of primary series schedules, including the most challenging 6, 10, 14 week (whether or not a birth dose of hepatitis B vaccine is given) and 2, 3, 4 month schedules, starting early in life and with 1-month intervals between doses.

Since initial registration, new clinical data have become available. Antibody persistence up to 4.5 years of age has been demonstrated for each antigen, when a second booster dose for diphtheria, tetanus, pertussis is recommended by WHO [1–3]. Longer term antibody persistence to 9–10 years of age was evaluated for HB, with a strong anti-HB response demonstrated even in subjects with antibody titers below seroprotective levels, indicating good priming and induction of persisting immune memory.

Post-licensure studies have confirmed the possible co-administration of Hexaxim with meningococcal vaccines (MenC or MenACWY conjugate vaccine), which is also included in several National Immunization Programs; data on co-administration with MenB vaccine will be available soon and will be of interest since many countries have introduced immunization against MenB in their infant programs.

While the initial clinical development plan focused on healthy term infants, post-licensure studies have provided data regarding use of Hexaxim in specific populations including preterm infants, immunocompromised infants or infants born to Tdap vaccinated women.

Real-world data have confirmed the effectiveness of Hexaxim to provide protection against pertussis to infants and toddlers, with VE that was comparable to a wP vaccine [55]. Throughout the clinical trials, Hexaxim has shown a favorable safety profile with no clinically meaningful differences compared to licensed vaccines.

Post-marketing safety monitoring is crucial to detect and evaluate any safety concerns temporally associated with vaccination. Since its launch in 2013, the safety profile of Hexaxim has been continuously monitored through routine pharmacovigilance with spontaneous AE reporting. Despite several limitations, such as under-reporting, differential reporting, or the maturity of pharmacovigilance systems, passive surveillance is likely to identify events and safety concerns at the population level. Post-marketing surveillance has confirmed that the favorable safety profile of Hexaxim and the frequency and the nature of AEs after Hexaxim vaccination are comparable to those in the product information. In addition, the reporting rates for anaphylactic reactions, extensive limb swelling, convulsion with or without fever, and HHE have been stable over the last 7 years and are in line with the published data for another hexavalent vaccine [99]. The O/E analyses performed for SIDS showed that the number of SIDS reported after Hexaxim vaccination was lower than the expected number in this population, which is in line with another hexavalent vaccine [99].

As a ready-to-use vaccine that includes aP and IPV antigens, Hexaxim is important in improving compliance to childhood vaccination schedules, in contributing to polio eradication, and is associated with a reduced incidence of vaccine administration errors since reconstitution is not required. Additionally, its use results in improved parental and healthcare worker satisfaction.

4. Conclusion

Over a period of approximately 20 years, an extensive program of clinical trials combined with continuous post-marketing surveillance have consistently shown a favorable safety profile and high immunogenicity of Hexaxim in a wide range of primary and booster vaccination schedules. As a fully liquid hexavalent vaccine, Hexaxim plays a critical role in the ongoing and future control of six pediatric infectious diseases globally.

5. Expert opinion

Pediatric combination vaccines are crucial for the continued control of a range of childhood infectious diseases. With the administration of several antigens in a single shot, such vaccines reduce the number of injections needed, improve compliance to increasingly complex vaccination schedules, and also improve operational efficiency.

Hexaxim is a hexavalent vaccine that builds on the success of precursor vaccines in the same family (including Tetraxim and Pentaxim) and provides protection against diphtheria, tetanus, pertussis, polio, hepatitis B, and invasive diseases caused by *Haemophilus influenzae* type b. High immunogenicity of each antigen following primary and booster administration has been demonstrated in a range of schedules including the Expanded Program on Immunization schedule, making it a suitable option for numerous National Immunization Programs. Moreover, the antibody persistence for each of the antigens up to school entry age, when children are to receive a booster dose of a diphtheria, tetanus, and pertussis-containing vaccine, is reassuring. The ability of Hexaxim to induce good priming and persisting immune memory against hepatitis B provides an alternative to separate administration of hepatitis B vaccine.

The efficacy and effectiveness of the acellular pertussis antigens contained in Hexaxim have been established and this vaccine has been shown to provide high levels of protection against pertussis disease in infants and toddlers.

While the clinical development program prior to licensure demonstrated a favorable safety profile, the post-marketing surveillance has continued to support vaccine safety following large-scale administration worldwide with more than 180 million doses of Hexaxim distributed since launch.

Recent data describe the use of Hexaxim as primary and booster vaccination in preterm infants which may represent a significant proportion of the birth cohort depending on the country. More limited data in immunocompromised infants are also available. In a context of greater adoption of pertussis vaccination during pregnancy, the good immunogenicity observed in infants born to Tdap vaccinated women is an asset.

As an established, fully liquid, ready-to-use, hexavalent vaccine, errors associated with reconstitution are avoided, thereby guaranteeing the delivery of all antigens to provide protection against each of the six diseases. Other advantages of Hexaxim are the inclusion of acellular pertussis and inactivated poliovirus antigens. Modern vaccination strategies include the switch from whole cell to acellular pertussis vaccines and from the use of oral poliovirus vaccine to inactivated poliovirus vaccine. The use of acellular pertussis antigens is associated with an improved safety profile, leading to better acceptance of vaccination and increased compliance to the vaccination schedule; the use of inactivated poliovirus vaccine antigens is important in the endgame strategy for the global eradication of polio.

In 5 to 10 years, the use of Hexaxim may be expected to increase the coverage for diphtheria-tetanus-pertussis vaccination and to improve the control of pertussis. Furthermore, within this timeframe, the use of Hexaxim may have contributed to the global eradication of polio. At 10 years after its first licensure, Hexaxim is already a cornerstone of protection against several childhood diseases worldwide, and in the future this is expected to be further strengthened.

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Declaration of interests

F Boisnard, C Manson, L Serradell, and D Macina are employees of Sanofi and may hold shares and/or stock options in the company. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or material discussed in the manuscript apart from those disclosed.

LIST OF ABBREVIATIONS

AE	adverse event
aP	acellular pertussis
DTaP-IPV	diphtheria, tetanus; acellular pertussis, inactivated poliovirus vaccine (Tetraxim, tetravalent vaccine, fully liquid)
DTaP-IPV//Hib	diphtheria, tetanus, acellular pertussis, inactivated poliovirus vaccine, and *Haemophilus influenzae* type b conjugate (Pentaxim, pentavalent vaccine, to be reconstituted)
DTPa-HBV-IPV/Hib	diphtheria, tetanus; acellular pertussis, hepatitis B, inactivated poliovirus vaccine, and *Haemophilus influenzae* type b conjugate (Infanrix hexa, hexavalent vaccine, to be reconstituted)
DTaP-IPV-HB-Hib	diphtheria, tetanus; acellular pertussis, inactivated poliovirus vaccine, hepatitis B, and *Haemophilus influenzae* type b conjugate (Hexaxim [or Hexyon, Hexacima, Hexarium depending on the country], hexavalent vaccine, fully liquid)
D	diphtheria
EMA	European Medicines Agency
EPI	Expanded Program on Immunization
EU	European Union
FHA	filamentous haemagglutinin
HB or Hep B	hepatitis B
HBsAg	hepatitis B surface antigen
HHE	hypotonic hyporesponsive episode
Hib	*Haemophilus influenzae* type b
HIV	human immunodeficiency virus
IPV	inactivated poliovirus vaccine
IU	International unit
Lf	limit of flocculation
LLOQ	lower limit of quantitation
MenACWY	MenACWY conjugate vaccine
Men B	meningococcal B vaccine
MenC	meningococcal C conjugate vaccine
O/E	observed to expected
OPV	oral poliovirus vaccine
PRN	pertactin
PRP	polyribosyl ribitol phosphate
PRP~T	*Haemophilus influenzae* type b capsular polyribosyl ribitol phosphate conjugated to tetanus toxoid
PT	pertussis toxin
RCT	randomized, controlled trial
SAE	serious adverse event
SIDS	Sudden Infant Death Syndrome
T	tetanus
Tdap	diphtheria, tetanus, pertussis (acellular component) vaccine (adsorbed, reduced antigen(s) content)
USA	United States of America
VAPP	Vaccine Associated Paralytic Polio
VDPV	Vaccine Derived Poliovirus
VE	vaccine efficacy
WHO	World Health Organization
wP	whole-cell pertussis

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A peer reviewer on this manuscript has disclosed that they have participated in hexavalent vaccine trials for Sanofi Pasteur, MSD and GSK and as an advisory board member for these companies. They have no shares of economic interest in these companies. All reviewers on this manuscript have received honoraria for their review work, reviewers on this manuscript have no relevant financial or other relationships to disclose.

Author contributions

All authors made substantial contributions to the development of the work, the analysis and interpretation of the data, reviewed it critically, approved the final version to be published, and are fully accountable for all aspects of the work.

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